Author: Oseko Ashpe

AFCON 2023 Final
Match Analysis Dashboard

Nigeria 1 – 2 Côte d'Ivoire · February 11, 2024
Stade Olympique Alassane Ouattara, Abidjan

1. Introduction

This report documents a football analytics project that was completed as a personal portfolio piece. The aim of the project was to build a professional match analysis dashboard for the 2023 Africa Cup of Nations (AFCON) Final, which took place on February 11, 2024, between Nigeria and Côte d'Ivoire at the Stade Olympique Alassane Ouattara in Abidjan. The final result was a 2–1 victory for Côte d'Ivoire.

The project was built entirely in Python using publicly available StatsBomb event data. The objective was to process the raw match event data and produce a single figure that summarised the key insight from the match across four visualisation panels: passing networks, shot maps, a match statistics table, and an expected goals (xG) flow chart.

2. Data Source

All data used in this project was sourced from StatsBomb Open Data, accessed via the statsbombpy Python library. StatsBomb provides free event-level data for selected competitions, including the full AFCON 2023 tournament. The match was identified using the following parameters:

Parameter	Value
Competition ID	1267
Season ID	107
Match ID	3923881

3. Tools and Libraries

The following Python libraries were used to complete the project:

Library	Purpose
statsbombpy	Loading and querying StatsBomb open event data

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mplsoccer	Drawing football pitches and parsing StatsBomb event structures via Sbopen
matplotlib	Building the figure, managing axes layout, and rendering all visual panels
pandas	Manipulating and filtering the event DataFrame throughout the pipeline
numpy	Performing numerical operations such as interpolation for node and edge sizing
Pillow	Image handling utilities

4. Methodology

4.1 Data Loading and Preprocessing

Match event data was loaded using sb.events() from statsbombpy and sorted by the event index column to preserve chronological order. The full dataset was then split into two team-specific DataFrames, one for Nigeria and one for Côte d'Ivoire, which were passed to each visualisation function independently.

4.2 Passing Networks

The passing network for each team was constructed inside a dedicated create_passnetwork() function. The focus was on: only completed, open-play passes made before the team's first substitution were included. This ensured that the network reflected the starting formation and structure rather than being distorted by personnel changes.

Each player was placed on the pitch at their average x/y position across all qualifying passes. Nodes were scaled by the number of passes made. Edges between pairs of players were drawn only where the pass count met a minimum threshold, and edge thickness and opacity were interpolated proportionally to pass frequency, with heavier lines indicating more frequent connections.

4.3 Shot Maps

Shot maps were drawn using the create_shotmap() function, which rendered each team's attempts on a vertical half-pitch using mplsoccer's VerticalPitch class. Each shot was plotted at its recorded x/y origin coordinate. The size of each marker was scaled by the StatsBomb xG value, so that larger circles represented higher-quality chances. Goals were filled with the team colour while non-goal attempts were shown as hollow circles with a coloured edge, allowing the viewer to distinguish between outcomes at a glance.

4.4 Match Statistics Table

A summary statistics table was produced by the create_table() function. For both teams, the following metrics were included in the table:

- Expected goals (xG
- Total shots
- Shots on target
- Total passes
- Pass completion percentage

4.5 xG Flow Chart

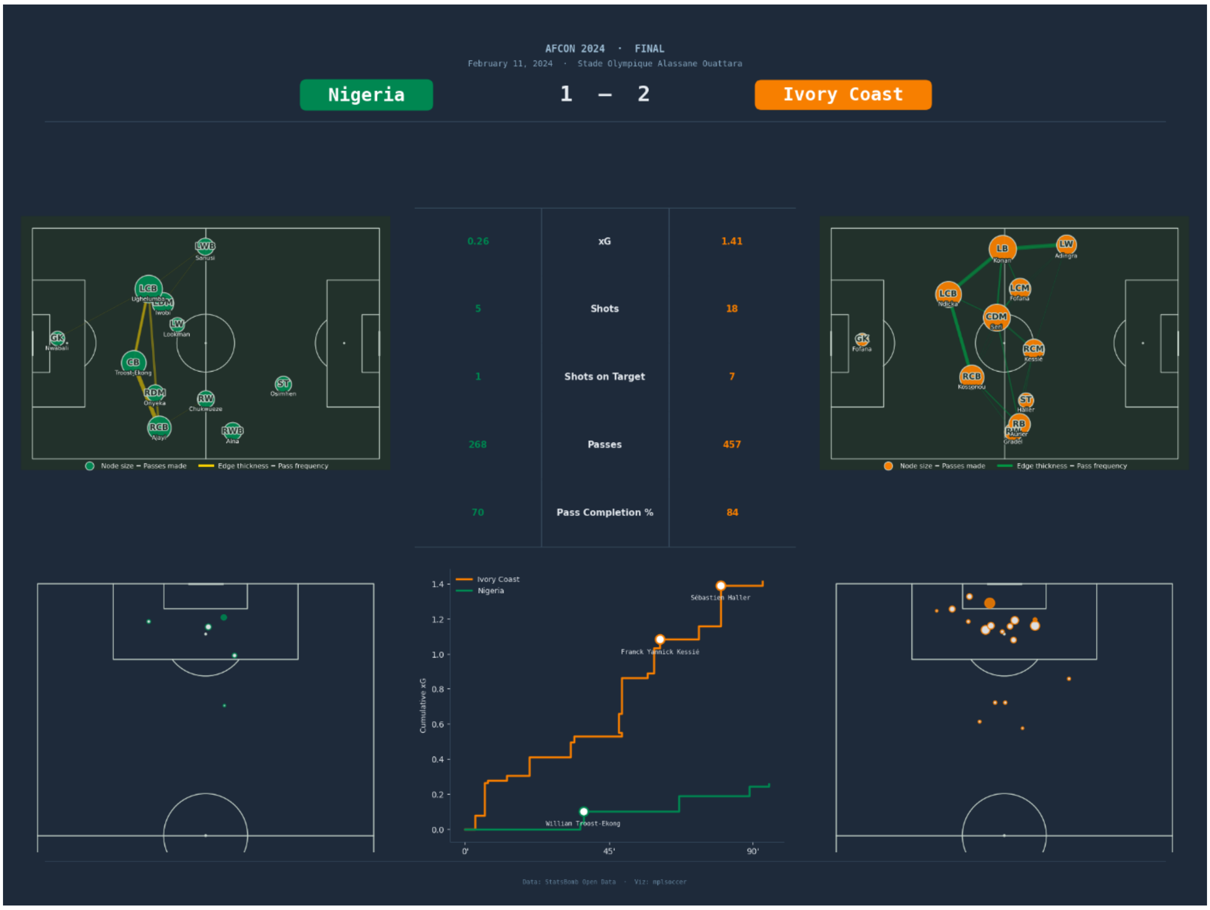
The xG flow chart was created by the create_xg_flow_chart() function. Shot events were sorted by match index, and cumulative xG was calculated for each team using a groupby() cumulative sum (cumsum) operation. The cumulative xG values were plotted against match minute, with goal events marked and labelled using the scorer's name. The x-axis was labelled at the 0-, 45-, and 90-minute marks.

4.6 Dashboard Assembly

The four panels were combined into a single 20 × 15 inch matplotlib figure. Each panel occupied a precisely positioned axes object added via fig.add_axes(), which allowed fine-grained control over placement without relying on a grid layout. The figure was structured as follows:

- A thin accent bar at the top, split between the two team colours
- A header section containing the competition label, date, venue, team name labels, and the scoreline
- An upper row containing the Nigeria passing network on the left, the match statistics table in the centre, and the Ivory Coast passing network on the right
- A lower row containing the Nigeria shot map on the left, the xG flow chart in the centre, and the Ivory Coast shot map on the right
- A footer with data and visualisation credits

The completed figure was saved as a PNG file at 150 dpi using plt.savefig(). The final dashboard produced is shown below:



5. Results

The dashboard successfully captured the key tactical and statistical story of the AFCON 2023 Final. Côte d'Ivoire won the match 2–1, with the xG flow chart showing that they created more and higher-quality chances throughout the match, particularly in the second half, where they overturned Nigeria's lead. Nigeria's passing network showed a compact, narrow structure, with the majority of connections concentrated through the central areas of midfield. Côte d'Ivoire's network, by contrast, demonstrated greater width and involvement from wide positions. The shot maps confirmed that Nigeria's attempts were concentrated in central areas in front of goal, while Côte d'Ivoire generated a greater volume of chances from a wider range of attacking positions. The match statistics table reinforced these findings, showing Côte d'Ivoire recorded higher xG, more shots, more shots on target, more passes, and a higher pass completion percentage.

6. Reflection

This project provided me with practical experience working with professional football event data and building a full analytics dashboard using Python. It involved transforming raw StatsBomb data into visualisations that communicate match insights clearly.

A key challenge was combining statsbombpy and the mplsoccer Sbopen parser, as both provide different parts of the event data. Aligning player and position information across the two sources required careful handling.

Another issue was the dashboard's visual design, where the original light background reduced readability. This was improved by switching to a dark background with light text, which significantly improved clarity.

Overall, the project strengthened my skills in football data analysis, data processing, and visual communication.